

Question				Marking details	Marks available																																			
					AO1	AO2	AO3	Total	Maths	Prac																														
2	(a)	(i)		correction 1 – (should be) acid (1) correction 2 – (should be pH) 7 (1) correction 3 – (should be) Na ₂ CO ₃ (1) award credit for correct corrections in any order	1 1	1		3		3																														
		(ii)		<table border="1"><thead><tr><th>Chemical Name</th><th>Chemical formula</th><th>Colour with Universal Indicator</th><th>pH</th><th>Acid, Alkali or Neutral</th></tr></thead><tbody><tr><td>Sulfuric acid</td><td>H₂SO₄</td><td>Green</td><td>1</td><td>Acid</td></tr><tr><td>Ethanoic acid</td><td>CH₃COOH</td><td>Orange</td><td>4</td><td>Alkali</td></tr><tr><td>Calcium hydroxide</td><td>Ca(OH)₂</td><td>Purple</td><td>14</td><td>Alkali</td></tr><tr><td>Water</td><td>H₂O</td><td>Green</td><td>5</td><td>Neutral</td></tr><tr><td>Sodium carbonate</td><td>NaCO₃</td><td>Blue</td><td>10</td><td>Alkali</td></tr></tbody></table> error 1 error 2 error 3 'green' circled do not accept more than one circled	Chemical Name	Chemical formula	Colour with Universal Indicator	pH	Acid, Alkali or Neutral	Sulfuric acid	H ₂ SO ₄	Green	1	Acid	Ethanoic acid	CH ₃ COOH	Orange	4	Alkali	Calcium hydroxide	Ca(OH) ₂	Purple	14	Alkali	Water	H ₂ O	Green	5	Neutral	Sodium carbonate	NaCO ₃	Blue	10	Alkali		1		1		1
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	(b)	(i)		carbon dioxide / CO ₂ (1) barium chloride / BaCl ₂ (1)	2			2		2																														
		(ii)		<u>limewater</u> goes <u>milky</u> accept alternative descriptions to milky e.g. turns white / cloudy	1			1		1																														

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	C		1		1		
			II	33		1		1	1	
		(ii)		exothermic	1			1		
	(d)	(i)		magnesium sulfate		1		1		
		(ii)		<u>lit splint</u> goes <u>pop</u> accept burning splint do not accept glowing splint	1			1		1
		(iii)	I	Zn ²⁺ and Cl ⁻	1			1		
			II	ZnCl ₂ do not allow ECF from part I		1		1		
				Question 2 total	8	6	0	14	1	8